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Infrastructure is Dead, Long Live Infrastructure in the Age of Agentic AI

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AGENDA

Session Overview

Infrastructure is Dead — Long Live Infrastructure in the Age of Agentic AI

Open (0–8 min) Setting the scene + opening poll

What Agentic AI actually means, and where your org sits today

Part 1 (8–22 min) The infrastructure paradigm shift

From manual pipelines to autonomous, intent-driven architectures

Part 2 (22–38 min) Agents in the DevOps lifecycle

Pipeline integration, live demo, guardrails and governance + Q&A

Part 3 (38–52 min) Real-world application

Two case studies, audience poll, and addressing your concerns

Part 4 (52–60 min) Getting started + open Q&A

The engineer's new role, adoption roadmap, key takeaways

Agentic AI | Azure | DevOps | Enterprise Cloud | Infrastructure

OPENING POLL

Where does your organisation sit on the cloud automation maturity curve?

Raise your hand for the option that best describes you today:

Level 1 — Manual

Infrastructure provisioned by hand; runbooks and tickets rule the day

Level 2 — Scripted

Terraform or Bicep in place, but humans drive every pipeline trigger

Level 3 — Automated

CI/CD pipelines running reliably; some self-healing in production

Level 4 — Intelligent

AI-assisted code review, anomaly detection, or auto-remediation in play

Poll | Audience Participation | Cloud Maturity

FOUNDATIONS

What is Agentic AI?

And what it definitely is not

A working definition

An Agentic AI system perceives its environment, reasons over a goal, plans a sequence of actions, executes those actions using tools, and adapts based on results — with minimal human intervention at each step.

Not a chatbot

Copilots suggest. Agents act. The difference is autonomy, memory, and tool use.

Not pure automation

Scripts follow fixed rules. Agents reason about context and handle novel situations.

Key capabilities that make it work

Goal decomposition, tool calling, memory, multi-step planning, and feedback loops

Agentic AI | Foundations | LLM | Azure AI Foundry

PART 1

The Old World

Manual pipelines, static configs, reactive operations

How infrastructure worked (and in many places still does)

Ticket-driven provisioning

Engineers open requests; wait times measured in hours or days; human bottlenecks at every stage

Static configuration drift

IaC templates describe desired state but nothing enforces it continuously; drift accumulates silently

Reactive incident response

Alerts fire, engineers wake up, they dig through logs, identify root cause, apply fix — repeat

Knowledge locked in individuals

Runbooks age out, tribal knowledge walks out the door, onboarding takes months

Legacy Ops | Manual Provisioning | Incident Response | IaC

PART 1

The New World

Autonomous agents, intent-driven infrastructure, proactive operations

What the AI-native infrastructure paradigm looks like

Intent-driven provisioning

An engineer describes the outcome in natural language; an agent reasons, generates IaC, validates, and deploys

Continuous compliance enforcement

Agents monitor configuration state in real time and remediate drift before it causes incidents

Proactive self-healing

Agents detect anomalies, correlate signals across services, and remediate — often before an alert fires

Institutional knowledge encoded

Runbooks become agent prompts; best practices are embedded in guardrails; expertise scales across teams

Agentic AI | Self-Healing | Intent-Driven | Azure

PART 1

Agentic AI Architecture on Azure

The building blocks available to you today

Azure AI Foundry

Model deployment, orchestration, and evaluation — the core runtime for your agents

Semantic Kernel

Microsoft's open-source SDK for building agent workflows that call Azure services as tools

GitHub Copilot for Azure

AI-assisted IaC generation, PR reviews, and pipeline troubleshooting inside your existing workflow

Azure DevOps + Pipelines

The CI/CD backbone that agents plug into to trigger, monitor, and validate deployments

Azure Policy + Defender for Cloud

The guardrail layer — agents can query, enforce, and remediate against policy definitions

Azure AI Foundry | Semantic Kernel | GitHub Copilot | Azure DevOps

PART 1

How Agents Interact with Azure

ARM, Bicep, Terraform, REST APIs — the agent toolbox

Tool calling as the integration mechanism

Agents don't have direct access to Azure — they call functions you expose. Each function wraps an Azure operation.

Infrastructure as code generation

Agent reasons about requirements, generates Bicep or Terraform, validates with what-if/plan, then executes via pipeline

ARM and REST API calls

For dynamic operations (scaling, restarting, tagging), agents call ARM REST endpoints directly with managed identity

Authentication and least-privilege

Agents authenticate as managed identities with scoped RBAC — no hardcoded credentials, full audit trail in Entra ID

ARM | Bicep | Terraform | REST API | Managed Identity

PART 2

Agents in the DevOps Lifecycle

Plan — Provision — Monitor — Remediate

Plan

Agent analyses requirements, estimates cost with Azure Pricing API, flags risks, proposes architecture

Provision

Agent generates IaC, runs validation (what-if/plan), awaits approval gate if configured, then triggers pipeline

Monitor

Agent continuously queries Azure Monitor, Log Analytics, and Cost Management; surfaces anomalies with context

Remediate

On detecting drift or incident, agent executes a known remediation or escalates to human with full context attached

DevOps | Azure Pipelines | Azure Monitor | Remediation

LIVE DEMO

Agentic Provisioning

Spinning up an Azure environment from a single natural language prompt

PART 2

Guardrails and Governance

Policy enforcement, approval gates, and audit trails

The trust problem — and how to solve it

Autonomous agents are only as trustworthy as the boundaries you put around them. Governance is not optional.

Azure Policy as the rulebook

Define allowed regions, SKUs, required tags, and network configurations. Agents check policy before acting.

Human-in-the-loop approval gates

For high-blast-radius actions (prod deployments, security group changes), agents pause and request explicit approval

Full audit trail in Entra ID and Monitor

Every agent action is logged with the managed identity, timestamp, and operation — indistinguishable from human audit trails

Azure Policy | Governance | Compliance | RBAC | Audit

Q&A — BREAK #1

Questions So Far?

5 minutes — anything from Parts 1 and 2

PART 3 — CASE STUDY

Autonomous Remediation

Multi-region Azure environment — persistent configuration drift

The scenario

A financial services client running workloads across Australia East, Southeast Asia, and West Europe was experiencing weekly configuration drift incidents. Manual detection averaged 4 hours; remediation added another 2.

The agentic approach

An agent was deployed to poll Azure Resource Graph every 15 minutes, compare live state against the approved Bicep baseline, and automatically remediate low-risk drift (tags, diagnostic settings, NSG rules) — escalating high-risk changes for human approval.

Case Study | Remediation | Azure Resource Graph | Multi-Region

The outcome

Mean time to detect dropped from 4 hours to under 2 minutes. 78% of drift events were remediated autonomously

PART 3 — CASE STUDY

Agent-Driven Cost Optimisation

Continuous right-sizing across a 400-VM estate

The scenario

A retail organisation running 400+ VMs across dev, test, and production environments was overspending by an estimated 35% due to over-provisioned SKUs and orphaned resources left running outside business hours.

The agentic approach

An agent was connected to Azure Cost Management and Monitor. It generated weekly right-sizing recommendations, automatically shut down dev/test resources outside business hours, and flagged orphaned disks and NICs for review.

Case Study | Cost Optimisation | Right-Sizing | Azure Cost Management

The outcome

Monthly Azure spend reduced by 28% within 60 days. Zero production incidents attributable to agent actions. The

POLL #2

What is your biggest concern about adopting agentic infrastructure?

Raise your hand for your top concern:

A — Security and blast radius

An agent with write access to production is a large attack surface if compromised

B — Auditability and compliance

How do I prove to auditors that agent actions meet our regulatory requirements?

C — Trust and explainability

I don't know why the agent made a particular decision — that makes me uncomfortable

D — Team readiness

We don't have the skills to build, run, or debug agentic systems yet

Poll | Audience Participation | Governance | Security

PART 3

Addressing the Concerns

Security, trust, blast radius, and compliance — head on

Security and blast radius

Scope agent identity to minimum required permissions. Separate identities per environment. Network-restrict agent egress.

Auditability and compliance

Every agent action via managed identity appears in Entra ID audit logs. Pair with Azure Monitor alerts for immediate visibility.

Trust and explainability

Require agents to produce a plan and rationale before execution. Log the reasoning chain. Implement mandatory human review for Tier 1 actions.

Security | Compliance | Trust | RBAC | Governance

Team readiness

Start with a single low-risk use case (e.g., dev environment provisioning). Build internal capability before expanding

PART 4

The Engineer as Orchestrator

From reactive administrator to strategic architect

The role is not disappearing — it is evolving

Engineers who thrive will be those who learn to design, direct, and govern agents — not those who resist the shift.

Skills that matter more

Prompt engineering for infrastructure, tool design, agent orchestration, policy definition, and outcome-based thinking

Skills that matter less

Remembering CLI flags, writing repetitive YAML, manually chasing alerts at 2am

The new value-add

Designing the guardrails, defining the goals, reviewing agent outputs, and escalating edge cases — humans set the intent, agents execute

Engineers | Orchestration | Skills | Future of Work

PART 4

A Practical Adoption Roadmap

Crawl — Walk — Run

Crawl — Low-risk, high-visibility wins (Months 1–3)

Deploy an agent for dev/test environment provisioning. Instrument with full logging. Let the team get comfortable with agent outputs.

Walk — Expand scope with human-in-the-loop (Months 3–6)

Add cost optimisation and drift detection agents. Introduce approval gates for production actions. Build a library of tested agent tools.

Run — Autonomous operations with governance layer (Month 6+)

Extend to production monitoring and remediation. Define clear escalation paths. Implement an agent health dashboard and anomaly detection on agent behaviour itself.

Adoption | Roadmap | DevOps | Governance | Azure

Guiding principle throughout

Start narrow, instrument everything, expand only what you trust, and always maintain a clear human override path.

KEY TAKEAWAYS

Infrastructure is Evolving,

Not Dying — Five things to take away today

1. **Agentic AI is an amplifier, not a replacement**

The best outcomes come from human intent + agent execution. Neither alone is sufficient.

2. **Azure has the building blocks today**

AI Foundry, Semantic Kernel, GitHub Copilot, and Azure Policy form a complete agentic stack right now.

3. **Guardrails are not optional**

Governance, least-privilege identity, and approval gates are what make autonomous systems trustworthy.

4. **Start small and instrument everything**

A single well-governed use case that builds team confidence is worth more than a grand vision that stalls.

5. **The engineer's role is expanding, not shrinking**

Orchestrating agents, designing guardrails, and encoding institutional knowledge is high-value, high-impact work.

Agentic AI | Azure | DevOps | Governance | Infrastructure

Q&A — OPEN DISCUSSION

Questions, Challenges, Ideas?

Let's talk — no such thing as a bad question

CONTACT

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